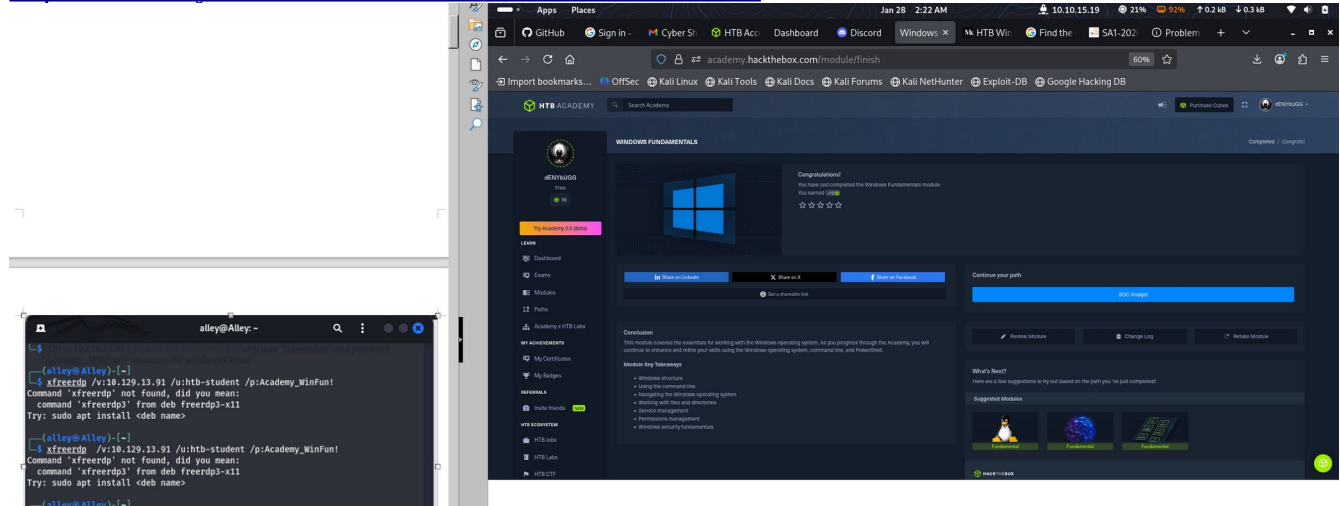


NAME: Alex Mbogo
GMAIL: evansnjiru5@gmail.com
ADMISSION NUMBER: ADC-CSS02-25064

WINDOWS FUNDAMENTALS

Module Completion Link

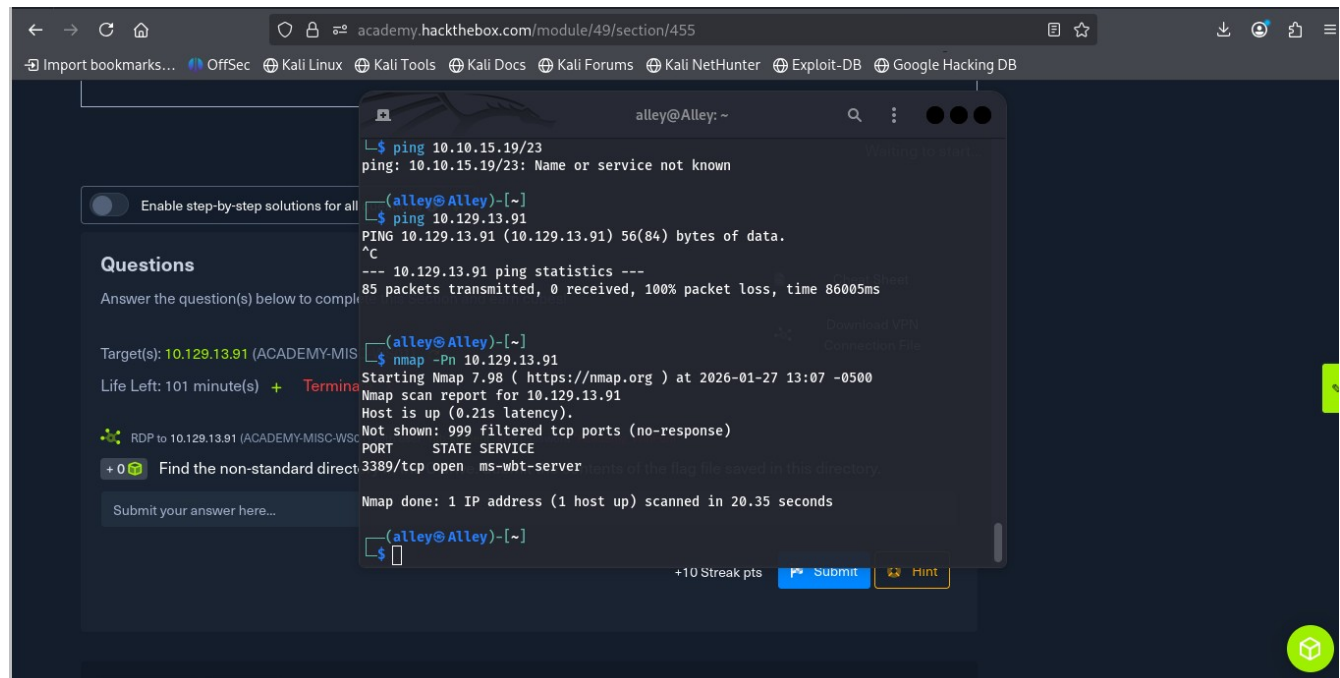
<https://academy.hackthebox.com/achievement/1378833/49>



1.1 Windows Operating system structure

Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

Scan for open ports with NMAP



Use the credentials to access target In my Decice

RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!" to access the windows Device

```
xfreerdp3 /v:10.129.13.203 /u:htb-student /p:Academy_WinFun!
```

The screenshot shows a web browser window with the URL `academy.hackthebox.com/module/49/section/455`. The page is titled "Waiting to start..." and features a toggle switch for "Enable step-by-step solutions for all questions". Below this, the "Questions" section displays the following information:

- Target(s): 10.129.13.91 (ACADEMY-MISC-WS01)
- Life Left: 97 minute(s) + Terminate X
- RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"
- +0 Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

At the bottom of the question section, there is a text input field labeled "Submit your answer here..." and two buttons: "Submit" and "Hint". The "Submit" button is highlighted with a blue border. The page also includes a "Cheat Sheet" link and a "Download VPN Connection File" link on the right side.



alley@Alley: ~



└─\$ RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!" to access the windows Device

└─(alley@Alley)-[~]

└─\$ xfreerdp /v:10.129.13.91 /u:htb-student /p:Academy_WinFun!

Command 'xfreerdp' not found, did you mean:

command 'xfreerdp3' from deb freerdp3-x11

Try: sudo apt install <deb name>

└─(alley@Alley)-[~]

└─\$ xfreerdp /v:10.129.13.91 /u:htb-student /p:Academy_WinFun!

Command 'xfreerdp' not found, did you mean:

command 'xfreerdp3' from deb freerdp3-x11

Try: sudo apt install <deb name>

└─(alley@Alley)-[~]

└─\$ sudo apt install freerdp3-x11 -y

[sudo] password for alley:

freerdp3-x11 is already the newest version (3.21.0+dfsg-1).

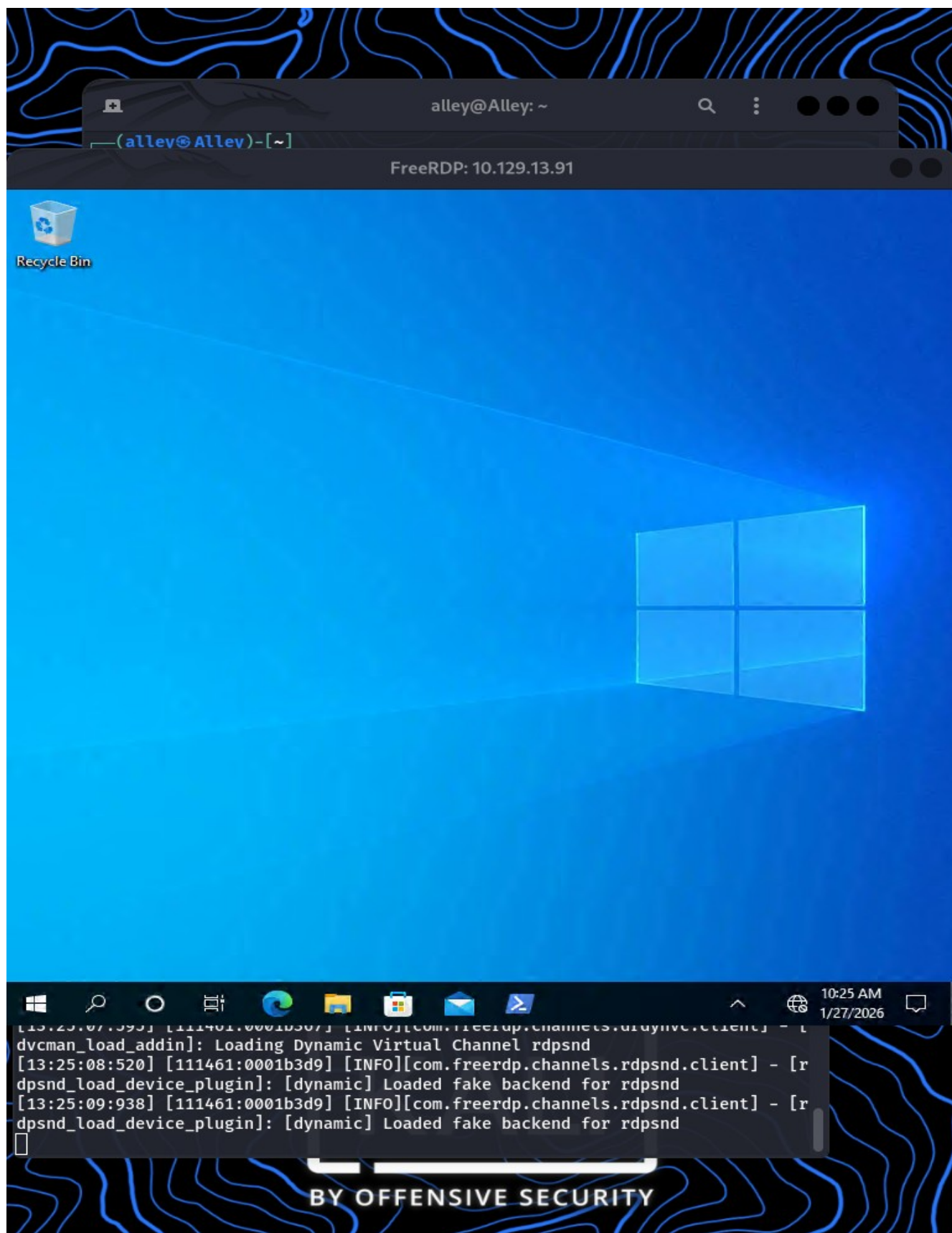
freerdp3-x11 set to manually installed.

Summary:

Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0

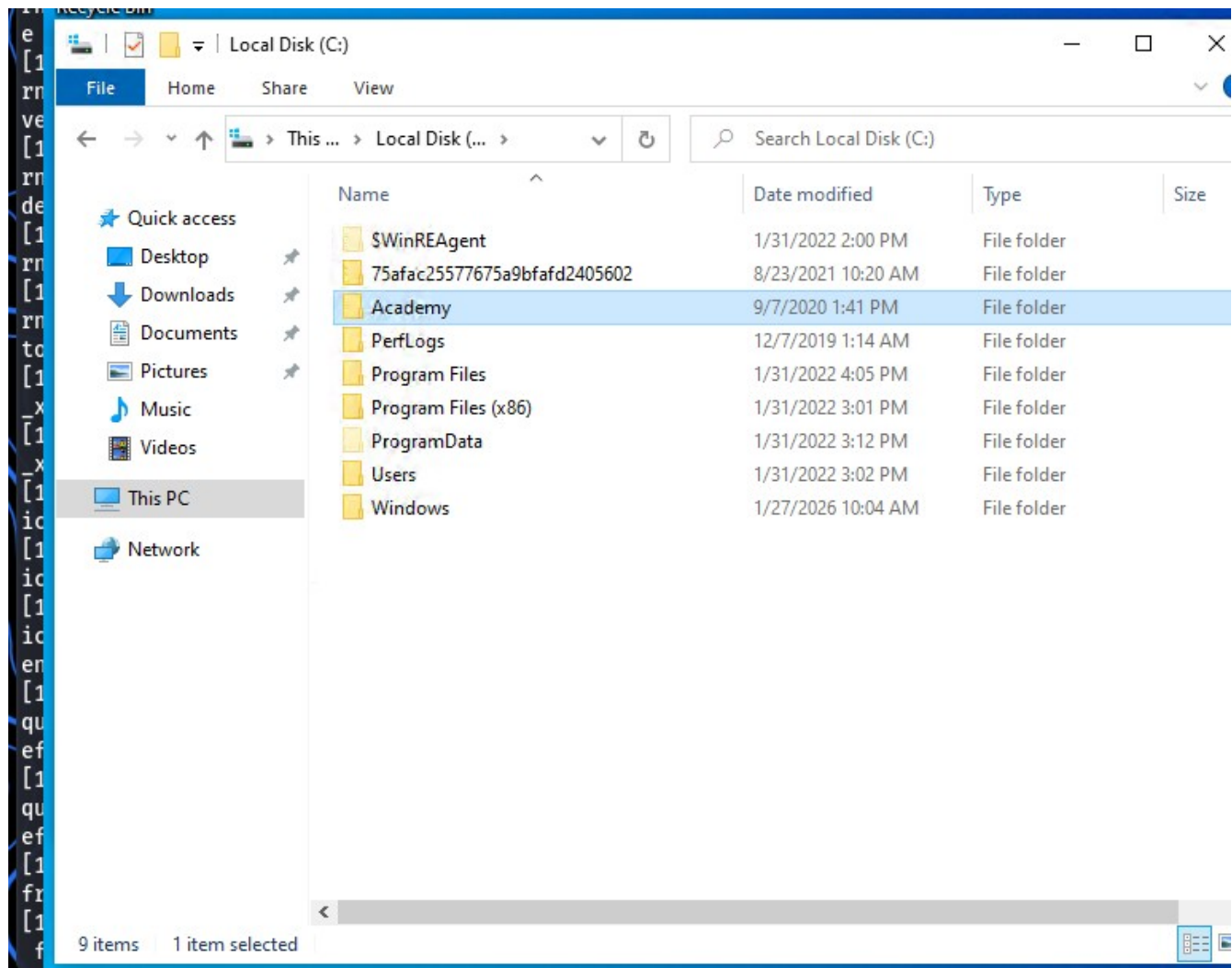
└─(alley@Alley)-[~]

└─\$ xfreerdp3 /v:10.129.13.91 /u:htb-student /p:Academy_WinFun!



connection successful

Here I look for the non-standard directory in the C drive



and then Proceed to identify the flag

FreeRDP: 10.129.13.91



Recycle Bin

Academy

File

Home

Share

View



<< Local Disk



Quick access



Desktop



Downloads



Documents



Pictures



Music



Videos



This PC



Network

1 item 1 item selected 32 bytes

flag - Notepad

File Edit Format View Help

c8fe8d977d3a0c655ed7cf81e4d13c75

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.13.91 (ACADEMY-MISC-WS01) 🔄

Life Left: 70 minute(s) + Terminate X

🔗 RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "hnb-student" and password "Academy_WinFun!"

+ 0 📦 Find the non-standard directory in the C drive. Submit the contents of the flag file saved in this directory.

c8fe8d977d3a0c655ed7cf81e4d13c75

Submit Hint

Previous Next

+10 Streak pts Mark Complete & Next

Cheat Sheet

Download VPN Connection File

New chat

Search chats

Images

Apps

Projects

GPTs

CV/Resume

Cyber Security

Explore GPTs

Your chats

xfreerdp not found

Kali Dev Cyber

Windows NT V

RDP Connection

Alex Evans

Free

Powered by HACKTHEBOX

File System

What system user has full control over The c:\users directory?

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.13.91 (ACADEMY-MISC-WS01) 🚀

Life Left: 32 minute(s) + Terminate X

🌟 RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "htb-student" and password "Academy_WinFun!"

+0 🧊 What system user has full control over the c:\users directory?

bob.smith

Submit

← Previous

Next →

+10 Streak pts

Mark Complete & Next

Cheat Sheet

Download VPN
Connection File

1 min

CPU1 11.2%

Memory and Swap

1 min
Memory
6.9 GB (83.7%) of 8.1 GB
Cache 2.9 GB

Network

1 min
Receiving
Total Received 699 MB

Disk

1 min
Reading
Total Read 0 MB

```
C:\Users\htb-student>icacls c:\users
c:\users: The system cannot find the file specified.
Successfully processed 0 files; Failed processing 1 files
```

```
C:\Users\htb-student>icacls c:/users
c:/users Everyone:(OI)(CI)(RX)
        NT AUTHORITY\SYSTEM:(OI)(CI)(F)
        BUILTIN\Administrators:(OI)(CI)(F)
        WS01\bob.smith:(OI)(CI)(F)
        BUILTIN\Users:(OI)(CI)(RX)
```

```
Successfully processed 1 files; Failed processing 0 files
```

```
C:\Users\htb-student>icacls c:\users_
```

NTFS vs. Share Permissions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.13.91 (ACADEMY-MISC-WS01) 🚀

Life Left: 24 minute(s) + Terminate X

🔗 RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "hib-student" and password "Academy_WinFun!"

+ 1 🗣️ What protocol discussed in this section is used to share resources on the network using Windows? (Format: case sensitive)

SMB

Submit Hint

+ 1 🗣️ What is the name of the utility that can be used to view logs made by a Windows system? (Format: 2 words, 1 space, not case sensitive)

Event Viewer

Submit Hint

Download VPN Connection File

Success
Congratulations! You earned 1 cube(s).

CPU1 6.1%

Memory and Swap

Memory 6.7 GB (81.4%) of 8.1 GB
Cache 3.0 GB

Network

Receiving 261 KB/s
Total Received 70 MB

Disk

Reading 0 KB/s

Windows Services & Processes

Identify one of the non-standard update services running on the host. Submit the full name of the service executable (not the DisplayName) as your answer.

FreeRDP: 10.129.13.91

Select Command Prompt

— Saved Pictures
— Saved Games
— Searches

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

Name	PID	Status	User name	CPU
SearchApp.exe	7148	Suspended	htb-student	00
RuntimeBroker.exe	6992	Running	htb-student	00
RuntimeBroker.exe	6508	Running	htb-student	00
RuntimeBroker.exe	7788	Running	htb-student	00
RuntimeBroker.exe	8048	Running	htb-student	00
RuntimeBroker.exe	880	Running	htb-student	00
RuntimeBroker.exe	676	Running	htb-student	00
Registry	100	Running	SYSTEM	00
rdpclip.exe	3528	Running	htb-student	00
notepad.exe	4136	Running	htb-student	00
notepad.exe	6788	Running	htb-student	00
nordvpn-service.exe	2768	Running	SYSTEM	00
NisSrv.exe	5172	Running	LOCAL SE...	00
MsMpEng.exe	3292	Running	SYSTEM	00
msdtc.exe	4340	Running	NETWORK...	00
MicrosoftEdgeUpdate.exe	5716	Running	SYSTEM	00
Microsoft.Photos.exe	1656	Suspended	htb-student	00
lsass.exe	720	Running	SYSTEM	00
LogonUI.exe	3500	Running	SYSTEM	00
LockApp.exe	7672	Running	htb-student	00
FoxitReaderUpdateService.exe	2108	Running	SYSTEM	00
fontdrvhost.exe	868	Running	UMFD-0	00

⬆ Fewer details

End task

Windows taskbar showing icons for Start, Search, Task View, Microsoft Edge, File Explorer, Mail, and others. System tray shows the time 11:33 AM and date 1/27/2026.

Waiting to start...

Success
Congratulations! You earned 0 cubes!

Enable step-by-step solutions for all questions

Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): 10.129.13.91 (ACADEMY-MISC-WS01)

Life Left: 17 minute(s) + Terminate X

RDP to 10.129.13.91 (ACADEMY-MISC-WS01) with user "hib-student" and password "Academy.WinFun!"

+ 0 Identify one of the non-standard update services running on the host. Submit the full name of the service executable (not the DisplayName) as your answer.

FoxitReaderUpdateService.exe

Submit Hint

Cheat Sheet

Download VPN Connection File

CPU1 25.7%

Memory and Swap

Memory 6.9 GB (83.7%) of 8.3 GB
Cache 3.0 GB

Network

Receiving 1.7 MB/s
Total Received 708 MB

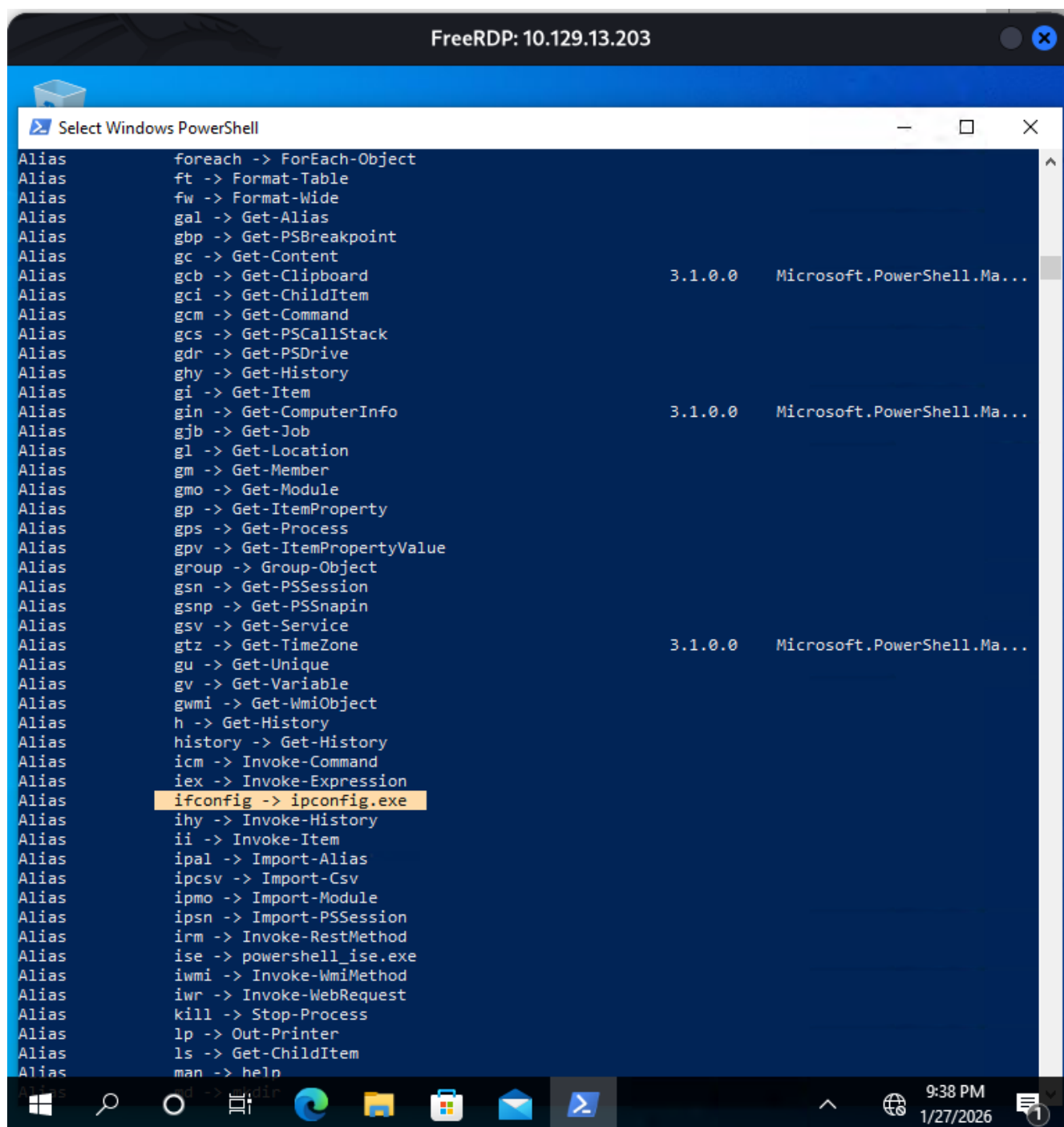
Disk

Reading 0.1 MB/s

Service Permissions

1). What is the alias set for the ipconfig.exe command?

ifconfig



2.) Find the Execution Policy set for the LocalMachine scope.

```
PS C:\Users\htb-student> get-Alias | where-Object { $_.Definition -eq "ipconfig.exe" }
```

```
PS C:\Users\htb-student> get-ExecutionPolicy -List
```

Scope ExecutionPolicy

MachinePolicy Undefined

Undefined

CurrentUser Undefined

Unrestricted

```
-----
UserPolicy
Process
Bypass
LocalMachine
```

```
FreeRDP: 10.129.13.203

Select Windows PowerShell

Alias      spps -> Stop-Process
Alias      spsv -> Stop-Service
Alias      start -> Start-Process
Alias      subj -> Suspend-Job
Alias      sv -> Set-Variable
Alias      swmi -> Set-WMIInstance
Alias      tee -> Tee-Object
Alias      trcm -> Trace-Command
Alias      type -> Get-Content
Alias      wget -> Invoke-WebRequest
Alias      where -> Where-Object
Alias      wjb -> Wait-Job
Alias      write -> Write-Output

PS C:\Users\htb-student> get-Alias -Name "ipconfig.exe"
get-Alias : This command cannot find a matching alias because an alias with the name 'ipconfig.exe'
does not exist.
At line:1 char:1
+ get-Alias -Name "ipconfig.exe"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (ipconfig.exe:String) [Get-Alias], ItemNotFoundException
+ FullyQualifiedErrorId : ItemNotFoundException,Microsoft.PowerShell.Commands.GetAliasCommand

PS C:\Users\htb-student> get-Alias | where-Object {$_.Definition -eq "ipconfig.exe" }
PS C:\Users\htb-student> get-Alias | Where-Object {$_.Definition -eq "ipconfig.exe" }
PS C:\Users\htb-student> get-Alias | where-Object { $_.Definition -eq "ipconfig.exe" }
PS C:\Users\htb-student> get-ExecutionPolicy -List

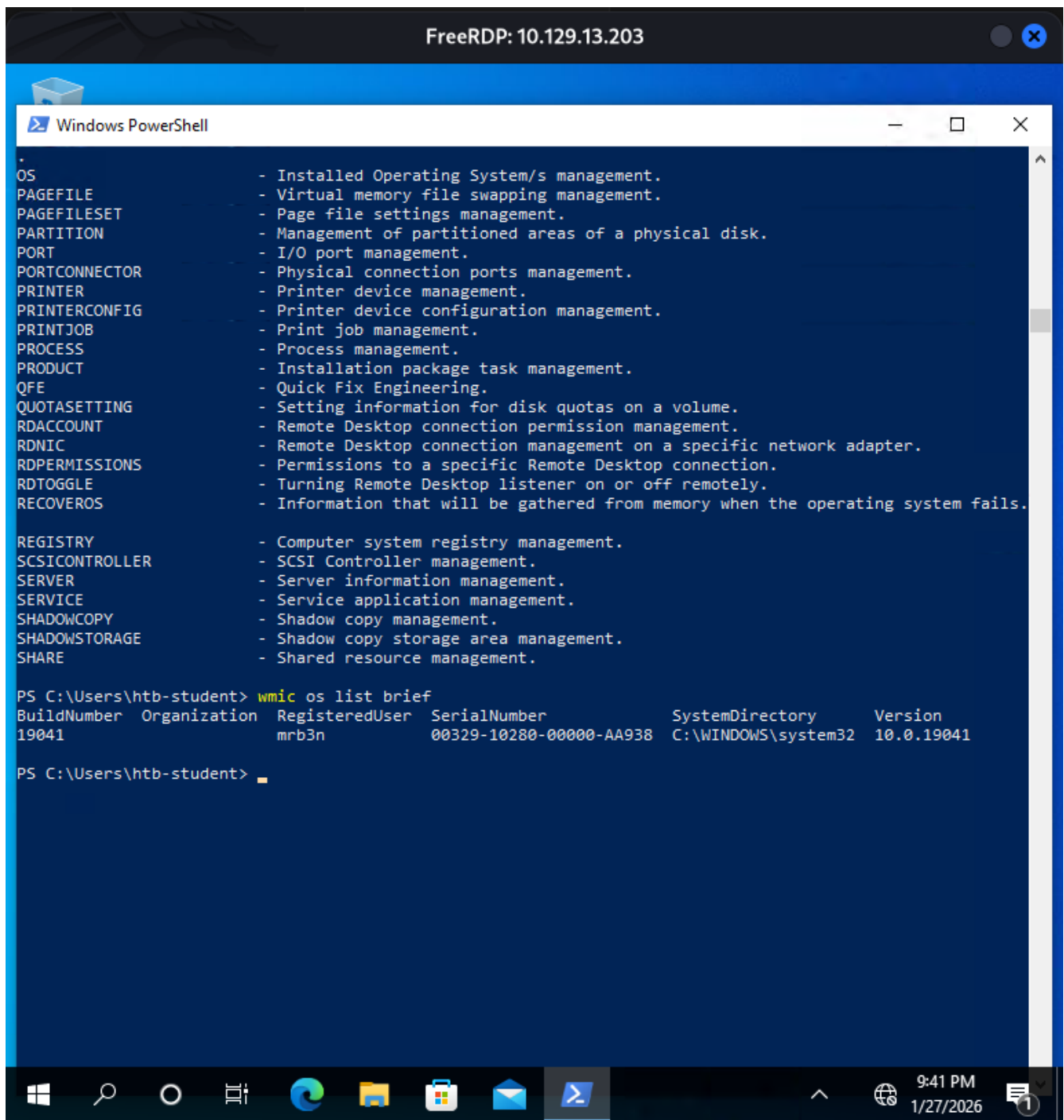
Scope ExecutionPolicy
-----
MachinePolicy Undefined
UserPolicy Undefined
Process Bypass
CurrentUser Undefined
LocalMachine Unrestricted

PS C:\Users\htb-student>
```

Windows Management Instrumentation (WMI)

A WMI provides system administrators with powerful tools for system monitoring

- 1). Use WMI to find the serial number of the system.



Microsoft Management Console (MMC)

The MMC can be used to group snap-ins, or administrative tools, to manage hardware, software, and network components within a Windows host.

Windows Subsystem for Linux (WSL)

WSL is a feature that allows Linux binaries to be run natively on Windows 10 and Windows Server 2019

Desktop Experience vs. Server Core

It is a minimalist server experience containing only key server services to reduce Attack surface. Server Core, all configuration and maintenance tasks are performed via the command-line, PowerShell, or remote management with MMC or Remote Server Administration Tools (RSAT).

Windows Security

1). Find the SID of the bob.smith user.

S-1-5-21-2614195641-1726409526-3792725429-1003

FreeRDP: 10.129.13.203

```
Windows PowerShell
PS C:\Users\htb-student> whoami /id bob.smith
ERROR: Invalid argument/option - '/id'.
Type "WHOAMI /?" for usage.
PS C:\Users\htb-student> Get-LocalUser -Name "bob.smith" |Format-List

AccountExpires      :
Description         :
Enabled             : True
FullName            :
PasswordChangeableDate : 9/7/2020 12:20:31 PM
PasswordExpires     :
UserMayChangePassword : True
PasswordRequired    : True
PasswordLastSet     : 9/7/2020 12:20:31 PM
LastLogon           :
Name                : bob.smith
SID                 : S-1-5-21-2614195641-1726409526-3792725429-1003
PrincipalSource      : Local
ObjectClass          : User

PS C:\Users\htb-student> Get-WmiObject Win32_UserAccount -Filter "Name='bob.smith'"

AccountType : 512
Caption     : WS01\bob.smith
Domain      : WS01
SID         : S-1-5-21-2614195641-1726409526-3792725429-1003
FullName    :
Name        : bob.smith

PS C:\Users\htb-student>
```

Taskbar: Windows Start button, Search, Task View, Edge, File Explorer, Mail, PowerShell, System tray (Network, Volume, Date/Time: 10:03 PM 1/27/2026, Notifications: 1).

2). What 3rd party security application is disabled at startup for the current user? (The answer is case sensitive).
NordVPN

Windows PowerShell

```
ReAccountType : 512
Caption       : WS01\bob.smith
Domain       : WS01
SID          : S-1-5-21-2614195641-1726409526-3792725429-1003
FullName     :
Name        : bob.smith

PS C:\Users\htb-student>
PS C:\Users\htb-student> GE-ItemProperty HKCU:\software\Microsoft\Windows\CurrentVersion\Run
GE-ItemProperty : The term 'GE-ItemProperty' is not recognized as the name of a cmdlet, function,
script file, or operable program. Check the spelling of the name, or if a path was included, verify
that the path is correct and try again.
At line:1 char:1
+ GE-ItemProperty HKCU:\software\Microsoft\Windows\CurrentVersion\Run
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (GE-ItemProperty:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\htb-student> Get-ItemProperty HKCU:\software\Microsoft\Windows\CurrentVersion\Run

OneDrive      : "C:\Users\htb-student\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background
NordVPN       : C:\Program Files\NordVPN\NordVPN.exe
PSPath        : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\software\Microsoft\Windows\CurrentVersion\Run
PSParentPath  : Microsoft.PowerShell.Core\Registry::HKEY_CURRENT_USER\software\Microsoft\Windows\CurrentVersion
PSChildName   : Run
PSDrive       : HKCU
PSProvider    : Microsoft.PowerShell.Core\Registry

PS C:\Users\htb-student>
```



10:08 PM
1/27/2026



Skills Assessment - Windows Fundamentals

Inlanefreight recently had an incident where a disgruntled employee in marketing accessed an internally hosted HR share and deleted several confidential files & folders. Thankfully, the IT team had good backups and restored all of the deleted data. There are now concerns that this disgruntled employee was able to access the HR share in the first place. After performing a security assessment, you have found that the IT team may not fully understand how permissions work in Windows. You are conducting training and a demonstration to show the department good security practices with file sharing in a Windows environment as well as viewing services from the command line to check for any potential tampering.

1. Creating a shared folder called Company Data

2. Creating a subfolder called HR inside of the Company Data folder

3. Creating a user called Jim

- Uncheck: User must change password at logon

FreeRDP: 10.129.13.203

Select Administrator: Windows PowerShell

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> New-LocalUser -Name "Jim" -Password (ConvertTo-SecureString "Password!" -AsPlainText -Force) -FullName "Jim User" -Description "Test User"

Name Enabled Description
----
Jim True Test User

PS C:\WINDOWS\system32> Set-LocalUser -Name "Jim" -PasswordNeverExpires $true
PS C:\WINDOWS\system32> New-LocalGroup -Name "HR" -Description "Security Group"

Name Description
----
HR Security Group

PS C:\WINDOWS\system32> Add-LocalGroupMember -Group "HR" -Member "Jim"
Add-LocalGroupMember : The term 'Add-LocalGroupMember' is not recognized as the name of
a cmdlet,
function, script file, or operable program. Check the spelling of the name, or if a path
was included,
verify that the path is correct and try again.
At line:1 char:1
+ Add-LocalGroupMember -Group "HR" -Member "Jim"
```

To change permissions, click Edit.

Edit...

Permissions for SYSTEM

Allow

Deny

Full control

✓

Modify

✓

Read & execute

✓

List folder contents

✓

Read

✓

Write

✓

For special permissions or advanced settings,
click Advanced.

Advanced

Close

Cancel

Apply



11:11 PM
1/27/2026

4. Creating a security group called HR

5. Adding Jim to the HR security group

6. Adding the HR security group to the shared Company Data folder and NTFS permissions list

- Remove the default group that is present
- Share Permissions: Allow Change & Read
- Disable Inheritance before issuing specific NTFS permissions
- NTFS permissions: Modify, Read & Execute, List folder contents, Read, Write

7. Adding the HR security group to the NTFS permissions list of the HR subfolder

- Remove the default group that is present
- Disable Inheritance before issuing specific NTFS permissions
- NTFS permissions: Modify, Read & Execute, List folder contents, Read, and Write

8. Using PowerShell to list details about a service

Assessment

What is the name of the group that is present in the Company Data Share Permissions ACL by default?
Everyone

What is the name of the service associated with Windows Update?
Get-Service

FreeRDP: 10.129.13.203

Select Administrator: Windows PowerShell

Stopped	WiaRpc	Still Image Acquisition Events
Running	WinDefend	Microsoft Defender Antivirus Service
Running	WinHttpAutoProx...	WinHTTP Web Proxy Auto-Discovery Se...
Running	Winmgmt	Windows Management Instrumentation
Stopped	WinRM	Windows Remote Management (WS-Manag...
Stopped	wisvc	Windows Insider Service
Stopped	WlanSvc	WLAN AutoConfig
Running	wlidsvc	Microsoft Account Sign-in Assistant
Stopped	wlpasvc	Local Profile Assistant Service
Stopped	WManSvc	Windows Management Service
Stopped	wmiApSrv	WMI Performance Adapter
Stopped	WMPNetworkSvc	Windows Media Player Network Sharin...
Stopped	workfolderssvc	Work Folders
Stopped	WpcMonSvc	Parental Controls
Stopped	WPDBusEnum	Portable Device Enumerator Service
Running	WpnService	Windows Push Notifications System S...
Running	WpnUserService_...	WpnUserService_64893
Running	wscsvc	Security Center
Running	WSearch	Windows Search
Running	wuauerv	Windows Update
Stopped	WwanSvc	WWAN AutoConfig
Stopped	XblAuthManager	Xbox Live Auth Manager
Stopped	XblGameSave	Xbox Live Game Save
Stopped	XboxGipSvc	Xbox Accessory Management Service
Stopped	XboxNetApiSvc	Xbox Live Networking Service

PS C:\WINDOWS\system32>
PS C:\WINDOWS\system32>

To change permissions, click Edit.

Edit...

Permissions for SYSTEM

	Allow	Deny
Full control	✓	
Modify	✓	
Read & execute	✓	
List folder contents	✓	
Read	✓	
Write	✓	

For special permissions or advanced settings, click Advanced.

Advanced

Close

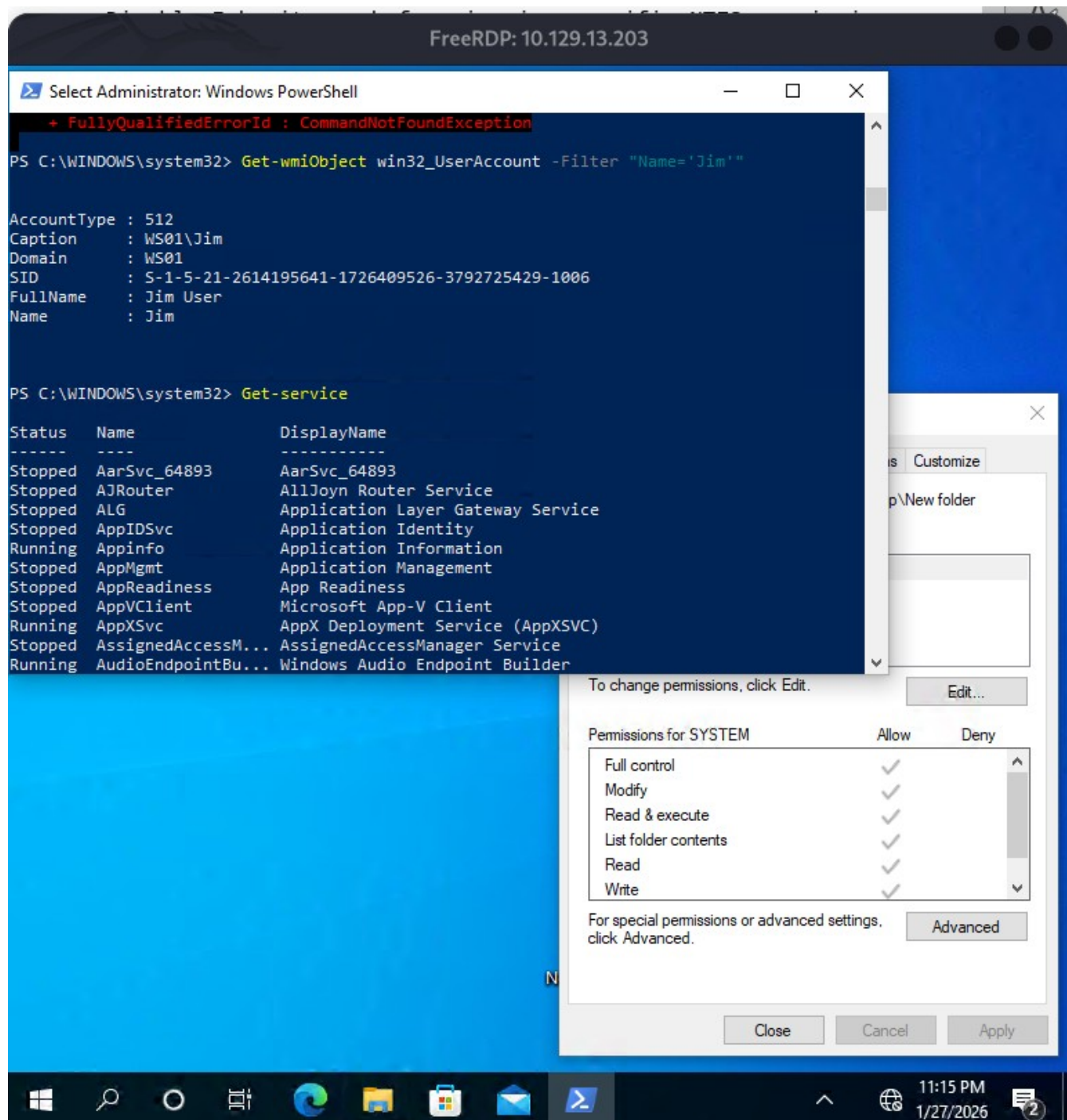
Cancel

Apply



11:14 PM
1/27/2026

List the SID associated with the user account Jim you created.



List the SID associated with the HR security group you create

FreeRDP: 10.129.13.203

Select Administrator: Windows PowerShell

```
+ CategoryInfo          : InvalidType: (:) [Get-WmiObject], ManagementException
+ FullyQualifiedErrorId : GetWMIManagementException,Microsoft.PowerShell.Commands.
GetWmiObjectCommand

PS C:\WINDOWS\system32> Get-LocalGroup -Name "HR" | Select Name, SID
>>
>> ^C
PS C:\WINDOWS\system32> Get-wmiObject win32_UserGroup -Filter "Name='HR'"
Get-wmiObject : Invalid class "win32_UserGroup"
At line:1 char:1
+ Get-wmiObject win32_UserGroup -Filter "Name='HR'"
+ ~~~~~
+ CategoryInfo          : InvalidType: (:) [Get-WmiObject], ManagementException
+ FullyQualifiedErrorId : GetWMIManagementException,Microsoft.PowerShell.Commands.
GetWmiObjectCommand

PS C:\WINDOWS\system32> Get-LocalGroup -Name "HR" | Select Name, SID

Name SID
----
HR    S-1-5-21-2614195641-1726409526-3792725429-1007

PS C:\WINDOWS\system32>
```

To change permissions, click Edit.

Edit...

Permissions for SYSTEM	Allow	Deny
Modify	✓	
Read & execute	✓	
List folder contents	✓	
Read	✓	
Write	✓	
Special permissions		

For special permissions or advanced settings, click Advanced.

Advanced

Close Cancel Apply

11:17 PM
1/27/2026

+ 1  What is the name of the group that is present in the Company Data Share Permissions ACL by default?

Everyone

 Submit

 Hint

+ 1  What is the name of the tab that allows you to configure NTFS permissions?

Security

 Submit

 Hint

+ 1  What is the name of the service associated with Windows Update?

WuauServ

 Submit

+ 1  List the SID associated with the user account Jim you created.

S-1-5-21-2614195641-1726409526-3792725429-1006

 Submit

 Hint

+ 1  List the SID associated with the HR security group you created.

S-1-5-21-2614195641-1726409526-3792725429-1007

 Submit

 Hint